Interpretable Control for Unstable Systems via SINDy-RL

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Abstract

Safety-critical autonomous systems require controllers that can be formally verified, audited, and deployed on resource-constrained hardware — properties that large neural networks cannot satisfy. SINDy-RL [1] addresses this by co-training a sparse surrogate world model and a neural policy in a Dyna loop, then distilling the result into a closed-form expression. We stress-test this pipeline on the inverted double pendulum (IDP) — a two-link system with two coupled unstable modes — asking whether it delivers its three core goals: data economy via the SINDy surrogate, a reduced-order policy via distillation, and interpretability via a sparse, auditable controller. Algorithm 1 treats the feature library Θ and the policy optimizer A as free practitioner inputs; our stress test spans both axes. After resolving two non-obvious engineering obstacles (polynomial degree ceiling and surrogate exploitation), we evaluate four library–optimizer combinations. A degree-3 polynomial library with PPO confirms data economy (66,277 real steps, **6.0× fewer** than the full-order baseline) and a reduced-order policy (82-term polynomial, 119× smaller than the baseline NN, 85% closed-loop success), but fails at interpretability: STLSQ retains 82 of 84 terms regardless of threshold — a consequence of ill-conditioning in the polynomial feature basis ($\kappa \approx 2.4 \times 10^4$), not a data or tuning problem. Replacing the polynomial library with 32 atoms derived from the IDP Euler-Lagrange equations — each physically meaningful by construction — eliminates the ill-conditioning at its source. Combined with SAC as the policy optimizer, this Lagrangian variant converges in 22,723 real steps (**17.6× fewer** than the baseline) with 100% task success and a 29-term distilled controller interpretable by construction: every retained term is a recognizable physical quantity. **SINDy-RL passes the stress test on the IDP.** With practitioner-appropriate choices of library and optimizer, the algorithm simultaneously achieves all three goals. The polynomial failure is a finding about representation selection, not a limitation of the framework.

1 Introduction

1.1 Motivation: The Inspectability Requirement

Deployed autonomous systems face a constraint that capable controllers must also be *inspectable*. Formal standards such as DO-178C (avionics software) and IEC 62443 (industrial control) require analyzable, auditable control laws [2], [3]; surgical robotics regulators may require that a control law be certifiable before permitting autonomous maneuvers near tissue; embedded actuators on spacecraft or small aerial vehicles have no floating-point stack capable of running a neural network at control rates. A ten-thousand-parameter neural network — however capable — offers no handle for stability proofs or formal verification, and its memory footprint alone disqualifies it from microcontroller deployment. A closed-form polynomial is the opposite: if sparse, each term has a physical interpretation; stability arguments can be constructed analytically, and an 84-term policy, while not necessarily interpretable, fits in kilobytes and evaluates as a single dot product.

This gap between what learning produces and what safety-critical systems can accept motivates a growing body of work on inherently interpretable controllers [3] — models whose structure is transparent by construction, not approximated after the fact.

1.2 SINDy and the Data Problem for Unstable Systems

Sparse Identification of Nonlinear Dynamics (SINDy [4]) offers a principled path to interpretable governing equations. Given a library of candidate functions over the state and input, sparse regression identifies which terms actually drive the dynamics and discards the rest. The resulting model is compact, physically grounded, and composed of a small number of terms that a practitioner can read and reason about. The difficulty is data. For an unstable equilibrium — such as an inverted pendulum — a random policy crashes in a handful of steps, and the near-upright transitions that SINDy needs are entirely unvisited. The system cannot provide the training data without a controller it does not yet have.

The resolution is to co-train the dynamics model and the controller iteratively. Sutton's Dyna architecture

[5] alternates between training a policy in a learned surrogate model and collecting new data from the real environment, so each iteration improves both components. Zolman et al. [1] apply this principle to SINDy by using an ensemble SINDy model as the Dyna surrogate, yielding SINDy-RL — a Reinforcement Learning (RL) framework that bootstraps the data problem while retaining interpretability as a downstream option.

1.3 Goals and Contributions

We stress-test SINDy-RL against its three core goals — (1) data economy via the SINDy surrogate, (2) reduced-order policy via distillation, and (3) interpretability via sparsity of the distilled controller — on the inverted double pendulum (IDP), a demanding two-link benchmark with two coupled unstable modes and a narrow region of attraction. Our contributions are:

1. **SINDy-RL passes the stress test.** With practitioner-appropriate library and optimizer choices, the algorithm achieves all three goals simultaneously on the IDP. The SAC+Lagrangian variant converges in 22,723 real-environment steps — **17.6× fewer than the full-order PPO baseline** — with 100% task success and a 29-term distilled controller interpretable by construction.
2. **Goal 1 — Data economy confirmed across all variants.** Every library–optimizer combination substantially outperforms the full-order baseline: PPO+polynomial (6.0×), PPO+no-x (11.4×), SAC+polynomial (13.0×), PPO+Lagrangian (15.4×), SAC+Lagrangian (17.6×). Pushing PPO+polynomial harder — a stricter convergence criterion consuming 323,934 real steps — confirms that additional iterations cannot substitute for the right design choices.
3. **Goal 2 — Reduced-order policy confirmed.** The SAC+Lagrangian distilled controller is 29 terms — 336× smaller than the baseline NN (9,731 parameters) — with 100% task success. The PPO+polynomial controller (82 terms, 119×, 85% success) also confirms the reduced-order goal.
4. **Goal 3 — Interpretability requires a physics-informed library.** The degree-3 polynomial basis is ill-conditioned on the IDP ($\kappa \approx 2.4 \times 10^4$), and STLSQ retains 82 of 84 terms regardless of threshold — a representation issue, not a data or tuning problem. The 32-atom Lagrangian library, derived from the IDP Euler-Lagrange equations, delivers interpretability by construction: every retained term is a physically meaningful quantity. This finding generalizes to any mechanical system with a known Lagrangian.
5. **Diagnosis and resolution of two non-obvious engineering obstacles** (polynomial degree ceiling and surrogate exploitation) with practical safeguards applicable to other unstable benchmark systems.

1.4 Ethics and Safety Considerations

Benefits. Interpretable controllers have direct safety benefits: closed-form polynomial expressions can be analyzed for failure modes before deployment, admit Lyapunov-style stability certificates in certain regimes, and allow human engineers to audit the control law. SINDy-RL's data efficiency reduces wear and risk on real hardware during training — fewer physical interactions means fewer dangerous exploration failures. Compact controllers enable deployment on hardware that cannot run large neural networks, potentially bringing safe autonomous control to resource-constrained applications in healthcare, agriculture, and infrastructure.

Risks. Surrogate exploitation — where a policy finds action sequences that the polynomial model predicts as highly rewarding but that do not correspond to real physics — is a failure mode that can produce overconfident deployment decisions. A practitioner who observes high surrogate reward without validating on the real environment may incorrectly conclude the system is ready for deployment. Incomplete sparsity (our 84-term result is compact but not truly auditable in the way a five-term equation would be) creates a risk that the "interpretability" framing overstates what an engineer can actually verify. Finally, distribution shift between the training distribution and deployment conditions can cause polynomial controllers to fail catastrophically in states not covered by the training data — a risk that perturbation augmentation mitigates but does not eliminate.

Any deployment of a SINDy-RL-derived controller in a safety-critical application should include real-hardware validation across the full intended operating envelope, formal analysis of the polynomial's behavior at boundary conditions, and a fallback mechanism (e.g., a conservative linear controller) for states outside the validated region.

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2 Technical Background

2.1 The Testbed: Inverted Double Pendulum

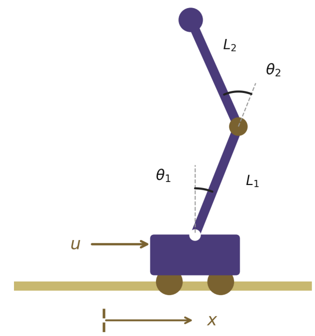

Figure 1. IDP geometry. State $\mathbf{x} = [x, \theta_1, \theta_2, \dot{x}, \dot{\theta}_1, \dot{\theta}_2]$. Tip height $h \in [0,\ 1.2]$ m; episode ends at $h \leq 1.0$ m.

The `InvertedDoublePendulum-v5` environment (MuJoCo 3.8.1 / Gymnasium 1.2.3) consists of two rigid links of equal length $L_1 = L_2 = 0.6$ m mounted on a sliding cart. The physical state is $\mathbf{x} = [x, \theta_1, \theta_2, \dot{x}, \dot{\theta}_1, \dot{\theta}_2] \in \mathbb{R}^6$, where x is the cart's horizontal position along the track, θ_1, θ_2 are joint angles measured from vertical, and dots denote time derivatives. The 9-dimensional observation replaces raw angles with sine/cosine encodings to avoid wrapping discontinuities. The single control input is a horizontal cart force $u \in [-1, 1]$.

Tip height $h = L_1 \cos\theta_1 + L_2 \cos(\theta_1 + \theta_2)$ reaches a maximum of 1.2 m when both poles are vertical. Gymnasium terminates an episode when $h \leq 1.0$ m, leaving only a 0.2 m near-upright band between success and failure. The per-step reward is:

$$r_k = 10 \cdot \mathbf{1} - (h_k - 2)^2 - 0.01\, x_{\text{tip}}^2 - \varepsilon \|\dot{\theta}\|^2$$

where the alive bonus (first term, ≈ 10/step) dominates when the system remains upright. Episodes are capped at 1,000 steps (50 s at $\Delta t = 0.05$ s).

2.2 SINDy-C: Sparse Dynamics Identification with Control

SINDy [4] identifies discrete-time dynamics by regressing the state increment against a polynomial library:

$$\mathbf{x}_{k+1} - \mathbf{x}_k = \underbrace{\Theta(\mathbf{x}_k,\, u_k)}_{\text{library}} \cdot \underbrace{\Xi}_{\text{sparse coefficients}}$$

For control-affine systems (SINDy-C [6]), the input u_k is included directly in the library.[1] The Sequentially Thresholded Least Squares (STLSQ) algorithm zeros coefficients below threshold λ, promoting sparsity in Ξ. A degree-d library over n variables contains $\binom{n+d}{d}$ terms; for the IDP's 7-dimensional state-action vector, degree-2 gives 36 features and degree-3 gives 120, a distinction that proved critical to convergence (§4.2).

2.3 E-SINDy: Ensemble Uncertainty Quantification

A single SINDy model provides a point estimate with no uncertainty information. Fasel et al. [7] address this with Ensemble SINDy (E-SINDy): fit M independent SINDy models on 80% bootstrap subsamples of the data, then at inference time report the mean and standard deviation of predictions across the ensemble. For $M = 10$ models, at each surrogate step `predict(x, u)` returns $(\mu_\Delta, \sigma_\Delta)$, where μ_Δ is the ensemble-mean predicted state increment and σ_Δ is the per-component standard deviation across members. High σ_Δ signals extrapolation beyond the training distribution. Following Zolman et al. §3.5 [1], we convert this into an active penalty: surrogate reward is reduced by $\kappa \cdot \text{mean}(\sigma_\Delta)$ per step ($\kappa = 5.0$), steering PPO away from high-uncertainty states.

2.4 Dyna-Style Model-Based RL and Behavioral Cloning

The Dyna architecture [5] alternates cheap model-based rollouts inside a learned surrogate with real-environment data collection. In SINDy-RL [1], the surrogate is the E-SINDy ensemble and the planner is PPO [8]. Figure 2 shows the RL control loop; in SINDy-RL the environment is instantiated twice: as the E-SINDy surrogate for cheap policy training, and as real MuJoCo for data collection and evaluation.

A Schroeder multi-sine sweep [9] bootstraps the initial dataset $\mathcal{D}$. Each Dyna iteration refits E-SINDy on near-upright transitions, runs PPO for 100k surrogate steps (warm-started from the prior policy), then collects 4,000 real transitions. After convergence, a selected teacher checkpoint is distilled via behavioral cloning:

[1] A system is control-affine if the control input appears linearly in the dynamics: $\dot{\mathbf{x}} = f(\mathbf{x}) + g(\mathbf{x})\, u$, where f and g may be arbitrarily nonlinear in the state. Most mechanical systems driven by forces or torques, including the IDP, satisfy this property.

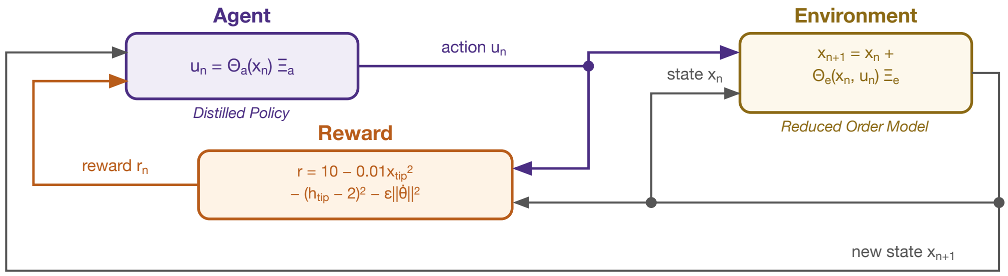

Figure 2. The RL control loop. The agent outputs action $u_k = \pi_\phi(\mathbf{x}_k)$ from the neural policy π_ϕ during Dyna training; after distillation into a polynomial, this becomes $u_k \approx \Theta_{\text{obs}}(\mathbf{x}_k)\,\xi$ where ξ is the sparse coefficient vector. In SINDy-RL, the environment is instantiated as the E-SINDy polynomial surrogate for cheap policy training and as the full MuJoCo simulator for real data collection and evaluation.

$$\min_{\Xi} \left\|\Theta_{\text{obs}}(X)\,\Xi - U^*\right\|_2 \quad (\text{STLSQ}, \ \lambda = 0.10)$$

where X is a matrix of observations, U^* are the corresponding NN policy actions, and Θ_{obs} is the degree-3 polynomial library over the 6-dimensional raw physical state $\mathbf{x} = [x, \theta_1, \theta_2, \dot{x}, \dot{\theta}_1, \dot{\theta}_2]$ (84 features, $\binom{9}{3} = 84$). Perturbation augmentation [10] (adding Gaussian noise to expert states and re-querying the trained neural network policy, referred to as the NN oracle) expands the 50k-transition dataset 5× to mitigate distribution shift without additional simulator rollouts. Figure 3 contrasts the two *distinct* libraries this involves: the 120-feature dynamics library the E-SINDy ensemble fits, and the 84-feature distillation library over the raw physical state the policy is cloned into.

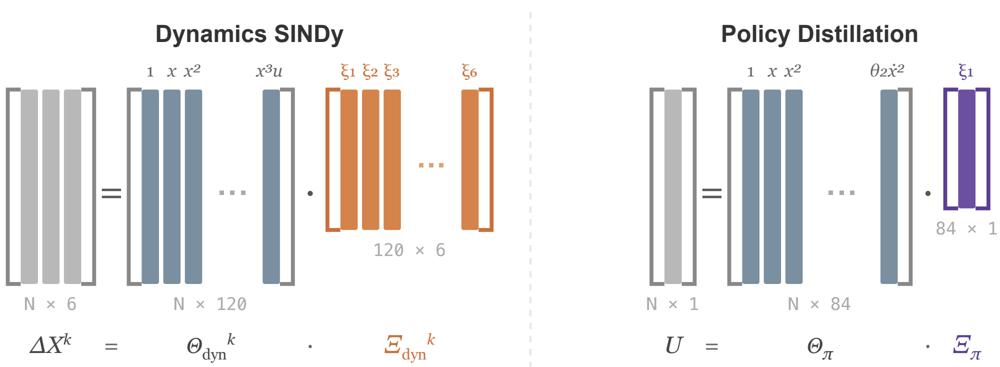

Figure 3. Sparse regression structure for E-SINDy dynamics identification (*left*) and policy distillation (*right*). Each of the $k = 1, \ldots, M$ ensemble members solves $\Delta\mathbf{X}^k = \Theta^k_{\text{dyn}}\Xi^k_{\text{dyn}}$, where $\Theta^k_{\text{dyn}} \in \mathbb{R}^{N\times 120}$ is the degree-3 polynomial library over the 7-dimensional state-action input evaluated on a bootstrap subsample, and $\Xi^k_{\text{dyn}} \in \mathbb{R}^{120\times 6}$ (orange) is the fitted coefficient matrix. Distillation fits a single $U = \Theta_\pi\Xi_\pi$, where $\Theta_\pi \in \mathbb{R}^{N\times 84}$ is the degree-3 library over the 6-dimensional raw physical state $[x, \theta_1, \theta_2, \dot{x}, \dot{\theta}_1, \dot{\theta}_2]$ ($\binom{9}{3} = 84$ features), and $\Xi_\pi \in \mathbb{R}^{84\times 1}$ (purple) is the scalar action coefficient vector. The two libraries are distinct: different input spaces (raw state-action vs. raw physical state) and different feature counts (120 vs. 84). Annotations below each Ξ give the nonzero coefficient count after STLSQ thresholding — 690 of 720 possible entries for the dynamics model and 82 of 84 for the distilled policy.

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3 Methods

The evaluation has three components. First, a full-order PPO agent trained with unlimited simulator access establishes the performance ceiling. Second, the SINDy-RL Dyna pipeline co-trains an E-SINDy surrogate and a PPO policy using a fraction of those real interactions, targeting comparable task success. Third, behavioral cloning — fitting a sparse polynomial to imitate the neural policy's actions, a process we refer to as *distillation* — converts the result into a closed-form expression suitable for embedded deployment and formal analysis. All three are implemented on the IDP testbed (§2.1).

3.1 Baseline: Full-Order PPO

The performance ceiling is a standard PPO agent (Stable-Baselines3 2.8.0) trained with unlimited real-environment access: a two-hidden-layer [64,64] MLP with tanh activations (9,731 parameters), trained for 400,000 total steps across 15,103 episodes. This agent is evaluated only as a reference point; the Dyna loop produces its own neural policy, and it is that policy — not the baseline — that is subsequently distilled into a polynomial. Baseline hyperparameters follow Zolman et al.'s defaults; characterizing the minimum viable architecture and step count — and thus the tightest fair comparison — remains future work.

3.2 SINDy-RL Pipeline

All code is implemented in Python 3.12.7 using PySINDy 2.1.0 [11] (E-SINDy surrogate), Stable-Baselines3 2.8.0 [12] (PPO policy), Gymnasium 1.2.3 with MuJoCo 3.8.1 [13] (simulation environment), NumPy 2.4.6 (numerical operations), and scikit-learn 1.8.0 (polynomial feature generation). The full pipeline is in `notebooks/sindy-rl.ipynb`.

Bootstrap (Stage 1). 300 episodes of Schroeder multi-sine excitation collect 2,897 state-transition pairs from real MuJoCo. Of these, 2,525 (87%) have tip height $h > 1.10$ m and are retained for SINDy fitting; the remaining 372 fall below the threshold and are discarded. Figure 4 shows the angular coverage: transitions are confined to a band of $|\theta_1| \lesssim 48°$ and $|\theta_2| \lesssim 67°$ around the upright equilibrium.

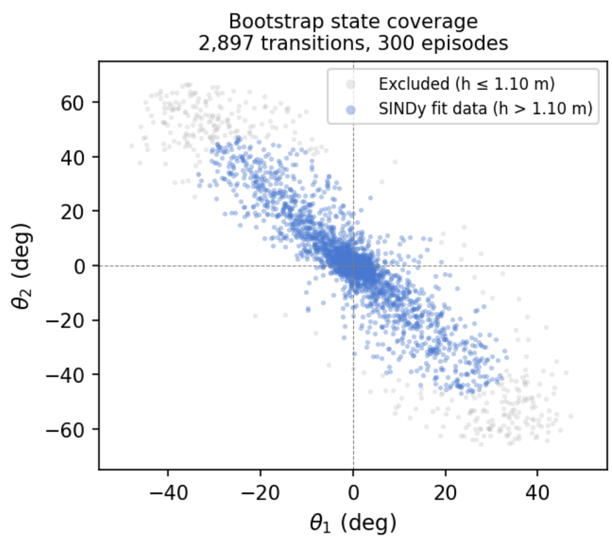

Figure 4. Bootstrap state coverage: 2,897 transitions from 300 Schroeder episodes plotted in joint-angle space. Blue points ($h > 1.10$ m, 87%) are retained for E-SINDy fitting; grey points are excluded. The Schroeder sweep keeps the system near vertical — the same near-upright operating domain the stabilizing policy will inhabit — so the surrogate need only be accurate in this small region of state space, not globally. Near upright ($\theta \approx 0$), the IDP dynamics are well-approximated by a degree-3 Taylor expansion; the inter-modal coupling terms that degree-2 cannot express (e.g., $\cos\theta_1 \cdot \cos\theta_2 \cdot \dot{\theta}_1$) appear at cubic order and are the leading nonlinear correction. This is why a degree-3 polynomial achieves RMSE $= 0.016$ from the first iteration despite the IDP being globally nonlinear.

E-SINDy fit (Stage 2). Transitions with tip height $h > 1.10$ m (poles within $\approx 24°$ of vertical) are retained.[2] Ten degree-3 SINDy-C models (PySINDy `SINDy` with `PolynomialLibrary(degree=3)` and `STLSQ(threshold=0.05)`) are fit on 80% bootstrap subsamples. Their coefficient matrices are stacked into

[2]The initial value was inherited from a reward-shaping constant `TIP_HEIGHT_TARGET = 2.0` (a dimensionless offset in MuJoCo's reward formula, not a physical height), setting `SINDY_H_MIN = 1.6` m — above the physical maximum $L_1 + L_2 = 1.2$ m. Every iteration silently fell back to fitting on all collected data until corrected to `SINDY_H_MIN = 1.10` m, derived from segment geometry.

a `FastEnsemblePredictor`[3] for efficient surrogate stepping.

Surrogate PPO (Stage 3). The predictor is wrapped in `EnsembleSurrogateEnv`[4], a Gymnasium environment that replicates MuJoCo's reward formula and $h \leq 1.0$ m termination condition, adds the uncertainty penalty $\kappa \cdot \text{mean}(\sigma_\Delta)$, and enforces physical state bounds ($|x| \leq 2.5$ m, $|\theta| \leq 0.9$ rad, $|\dot{\theta}| \leq 12$ rad/s). PPO trains for 100k steps; early-stopped if mean surrogate episode length stays below 5 steps after 50k steps.

Real data collection (Stage 4). The policy is deployed in real MuJoCo for 4,000 steps, appended to $\mathcal{D}$.

Repeat with warm-start (Stage 5). If exploitation is detected (surrogate reward $> 3\times$ previous AND real episode length $< 50\%$ of best seen), the next iteration rolls back to the best real-env checkpoint.

Distillation. 50k expert transitions are collected from the selected post-hoc teacher checkpoint in real MuJoCo, augmented 5× with per-dimension Gaussian noise (σ: [0.02, 0.02, 0.02, 0.05, 0.10, 0.10] for $[x, \theta_1, \theta_2, \dot{x}, \dot{\theta}_1, \dot{\theta}_2]$), and re-queried from the NN oracle. A degree-3 polynomial is then fit via STLSQ ($\lambda = 0.10$) on the 6-dimensional raw physical state $[x, \theta_1, \theta_2, \dot{x}, \dot{\theta}_1, \dot{\theta}_2]$ (84 features at degree-3, $\binom{9}{3} = 84$).

3.3 Metrics

Data efficiency is measured by real-environment step count. Task performance is measured by success rate (episodes lasting at least 500 steps) and mean episode length. Surrogate quality is tracked by E-SINDy one-step RMSE on held-out near-upright transitions. Distillation quality is measured by OLS R^2 and STLSQ term count.

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4 Results

4.1 Baseline

Full-order PPO achieves mean reward $9{,}324 \pm 2$, 100% success, and mean episode length 1,000/1,000 steps at a cost of 400,000 real simulator interactions and a 9,731-parameter opaque network. This is the performance ceiling.

4.2 Dyna Loop Convergence

The Dyna loop converged at iteration 11 using **66,277 real steps — 6.0× fewer than the baseline**. Performance degrades if the loop is continued past the convergence peak — a consequence of continued surrogate exploitation corrupting the dataset — motivating cross-validated best-checkpoint selection rather than using the final policy. Table 1 shows iterations 1–6 (39,616 cumulative steps), representing the early convergence trajectory; the loop continued to iteration 11. Cross-validated scanning of all saved checkpoints identified iteration 7 as the best performer: **90% success, mean episode length 904 steps**. The converged E-SINDy dynamics model is dense: 690 out of 720 possible coefficients (120 features × 6 state dimensions) are nonzero, indicating that this polynomial basis and STLSQ setting produced a near-dense model.

Iteration	Cumul. real steps	SINDy RMSE	Surr. mean len	Real mean len	Success
Bootstrap	2,897	0.016	—	—	—
1	7,015	0.016	11.8	12	0%
2	11,186	0.020	17.1	17	0%
3	15,551	0.084	36.5	36	0%
4	19,979	0.095	42.8	43	0%
5	29,453	0.085	547	547	50%

[3] The bottleneck was `PolynomialLibrary.transform()` from scikit-learn, which carries ~1 ms fixed overhead per call regardless of input size; calling it once per ensemble member cost ~10 ms/step and ~13 min per 75k-step PPO phase. The fix: at construction, extract the `powers_` exponent matrix from sklearn's `PolynomialFeatures` once; at each step, compute features as `np.prod(xu ** powers_, axis=1)` in pure NumPy and apply all 10 pre-stacked `(10, 6, 120)` coefficient matrices via a single batched matmul (~0.93 ms/step, 11.5× speedup).

[4] Gymnasium wrapper around the E-SINDy ensemble: reward = MuJoCo formula (§2.1) minus $\kappa \cdot \text{mean}(\sigma_\Delta)$; termination at $h \leq 1.0$ m or outside physical limits with a -50 out-of-bounds penalty. Matching MuJoCo's conditions exactly is necessary for policy transfer.

Iteration	Cumul. real steps	SINDy RMSE	Surr. mean len	Real mean len	Success
6	**39,616**	**0.080**	**616**	**616**	**60%**

Post-hoc checkpoint scan	Eval episodes	Real mean len	Success ($\geq$ 500 steps)
7	20	904	90%

The 66,277-step count covers the full 11-iteration Dyna run; Table 1 shows the first six iterations (39,616 cumulative steps). The iteration-7 checkpoint (90% success) is the cross-validated best within this run. The RMSE rise at iterations 3–4 reflects a better policy exploring states further from vertical, not model degradation; the surrogate remained accurate in the near-upright band.

Strict full-horizon stress test. To test whether the surrogate-trained PPO policy could match the baseline rather than merely reach the report's 500-step success criterion, we ran a stricter continuation experiment requiring all evaluation episodes to survive the full 999-step horizon. This is a harder metric than the 90% result reported above, not a direct repetition of the same success definition. The stricter loop used larger real-data corrections (6,000 transitions/iteration), 150k surrogate PPO steps/iteration, fixed seed-block evaluation, targeted failed-seed replay, and rollback on real-performance regression. This did not close the remaining gap. The best surrogate-only checkpoint reached **80% full-horizon success** and mean episode length 805 at iteration 12 after **108,251 counted real MuJoCo interactions**; continuing to 20 iterations consumed **323,934 interactions** without exceeding that peak, with later checkpoints oscillating between 40–80% success. A preliminary real-PPO fine-tune initialized from the best surrogate checkpoint added 57,344 real training steps and still evaluated at only **77% full-horizon success** on the 30-seed final evaluation. We therefore treat the original Dyna result as a data-efficient partial-success controller, not as a baseline-matching controller. This negative result is consistent with the failure mode emphasized by Zolman et al. [1]: Dyna-style SINDy-RL can be highly sample-efficient, but the neural policy may overfit or drift when the surrogate no longer captures the rollout distribution tightly enough.

4.3 Engineering Obstacles

Two non-obvious obstacles prevented convergence on early attempts.

Degree-2 RMSE ceiling. Over 25 iterations with data growing from 5k to 90k transitions, RMSE oscillated at 0.10–0.16 and real episode length grew from only 6 to 22 steps. A degree-2 library (36 features) cannot express the inter-modal coupling terms that dominate IDP dynamics (e.g., $\cos\theta_1 \cdot \cos\theta_2 \cdot \dot{\theta}_1$), which are cubic. When RMSE fails to decrease with 10× more data, the cause is model capacity, not data quantity. Fix: `SINDY_DEGREE=3` (120 features), which dropped RMSE to 0.013 within two iterations.

Surrogate exploitation. In a diagnostic run, surrogate reward jumped 9× (497 to 4,525) while real episode length collapsed 87% (414 to 56 steps). The policy found action sequences the polynomial predicted as highly rewarding that had no correspondence to real physics. Uncertainty penalization alone is insufficient: all 10 ensemble members share the same polynomial basis, so in extrapolated regions they all make the same wrong prediction simultaneously and ensemble disagreement σ_Δ remains low. Rollback alone is also insufficient: it detects exploitation after the fact but does not prevent the exploited iteration from degrading the dataset. Fix: both mechanisms together — uncertainty penalty `reward -= 5.0 * mean(sigma_delta)` during surrogate PPO, plus rollback to best real-env checkpoint when surrogate reward exceeds 3× the previous value and real episode length drops below 50% of best seen. The design principle generalizes: any ensemble surrogate built on a shared function basis cannot detect shared extrapolation errors through internal disagreement — real-environment feedback is the only reliable out-of-distribution signal.

4.4 Policy Distillation

Behavioral cloning from the best Dyna checkpoint (iteration 7, 90% success) produced a degree-3 polynomial with $R^2 = 0.911$. Five additional obstacles required resolution: (1) the distillation teacher must be the best-checkpoint policy, not the final loop policy, which may have drifted during surrogate training (using the final policy gave 0% success); (2) degree-2 gave $R^2 \approx 0.905$ regardless of data volume, requiring degree-3; (3) perturbation augmentation was needed to close distribution shift between the NN's training trajectories and deployment states; (4) the sin/cos observation encoding is ill-conditioned for polynomial regression near the upright operating point — when $\theta \approx 0$, $\cos\theta \approx 1$ so $\cos^2\theta \approx 1$, $\cos\theta_1 \cos\theta_2 \approx 1$, and so on, making many columns of Θ_π nearly identical, while the identity $\sin^2\theta + \cos^2\theta = 1$ injects exact linear dependences at every polynomial degree; switching to the raw 6-dimensional physical state $[x, \theta_1, \theta_2, \dot{x}, \dot{\theta}_1, \dot{\theta}_2]$ ($\binom{9}{3} = 84$ features) eliminates these collinearities and is the implementation used here; (5) the Dyna datastore provides only ∼28k near-upright transitions clustered tightly along the expert's closed-loop trajectory — behavioral cloning errors compound toward adjacent states that were never in the training distribution, causing runaway divergence; replacing datastore relabeling with surrogate rollout (running the best-checkpoint PPO inside the E-SINDy surrogate for exactly 50k steps) samples from the policy's actual visit distribution at zero additional real-environment cost and provides broader near-upright coverage.

STLSQ at $\lambda = 0.10$ retained **82/84 terms** (two terms dropped) with **85% closed-loop success** (Figure 5). The distilled policy is 119× smaller than the baseline NN. Figure 6 shows a threshold ablation across $\lambda \in [0.001, 1.0]$: success remains at 90–95% as the term count decreases, confirming that distillation performance is robust to the threshold choice. The dominant policy terms include recognizable physical structure: a constant bias; proportional angle feedback on θ_1 and θ_2; velocity damping on $\dot{\theta}_1$ and $\dot{\theta}_2$; and inter-pole coupling — the physically expected dominant nonlinearity. The remaining cubic cross-terms encode residual nonlinearity from the NN's tanh activations.

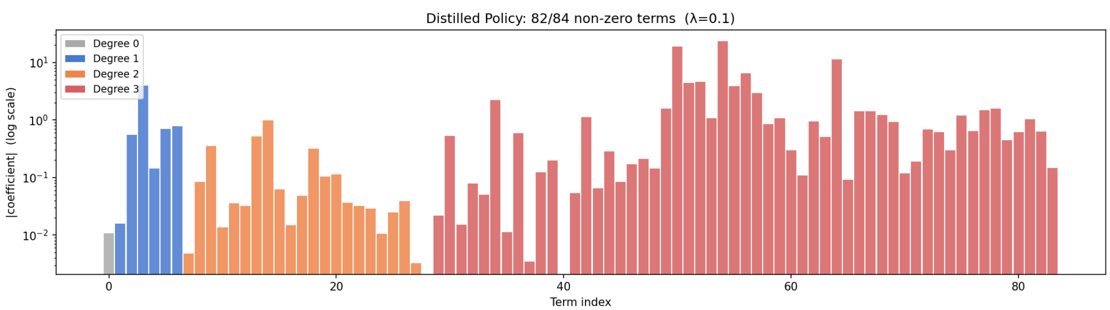

Figure 5. Distilled policy coefficient magnitudes (log scale), coloured by polynomial degree. 82 of 84 terms are retained at $\lambda = 0.10$. Distilled success is 85%.

The dynamics feature matrix Θ (209,620 × 120) has condition number $\kappa = 2.37 \times 10^4$ (full rank, 120/120 singular values nonzero; a five-decade spread between largest and smallest singular values). In an ill-conditioned feature basis, small OLS coefficients along low-singular-value directions are numerically unreliable; STLSQ cannot safely threshold any of the 82 retained terms without risking structural contributions. The 690/720 nonzero dynamics coefficients reflect the same phenomenon — not that all terms are mechanistically necessary, but that the feature basis does not admit a stable sparse decomposition. As on other unstable systems, $R^2 = 0.911$ on the distillation set is a necessary but not sufficient metric: small in-distribution approximation errors can compound against the IDP's unstable modes in closed loop.

Distillation preserves neural-policy performance while delivering a closed-form controller compatible with embedded hardware deployment (Goal 2, confirmed). The dense result (82/84 terms) is a property of the polynomial feature basis on this system — §4.6 shows that switching to a physics-informed Lagrangian basis achieves interpretability by construction.

Approach	Real-env steps	Mean ep len	Success (≥ 500 steps)	Params
Baseline PPO	400,000	1,000	100%	9,731
SINDy-RL NN (Dyna iter 6, early)	39,616	616	60%	9,731

Approach	Real-env steps	Mean ep len	Success ($\geq$ 500 steps)	Params
SINDy-RL NN (best checkpoint, iter 7)	within 66,277-step run	904	90%	9,731
SINDy-RL Polynomial (distilled from iter 7)	66,277 + 50,000†	904	85%	84 terms

$ˆ†$50,000 rollout steps for distillation data; 5× perturbation augmentation reuses the NN oracle without additional MuJoCo interactions. The 66,277-step run covers all 11 Dyna iterations; iteration 7 is the cross-validated best checkpoint within this run.

4.5 Physics-Informed Feature Reduction via Translational Symmetry

The IDP equations of motion are invariant under cart translations $x \to x + c$: the Lagrangian contains $\dot{x}$ but not x, so the state increment $\Delta x = \dot{x}\,\Delta t$ is given exactly by kinematics and need not be learned. Removing x from the SINDy library reduces the state-action input from 7 to 6 dimensions and cuts the degree-3 feature count from $\binom{10}{3} = 120$ to $\binom{9}{3} = 84$ — a 30% reduction — while hard-coding $\Delta x = \dot{x} \cdot \Delta t$ in the surrogate step. For the no-x variant, the distillation library also removes x from the policy input, using the 5-dimensional state $[\theta_1, \theta_2, \dot{x}, \dot{\theta}_1, \dot{\theta}_2]$ (56 features at degree-3, $\binom{8}{3} = 56$).

With the reduced library the Dyna loop converges in seven iterations using 35,111 real-environment steps. The converged neural policy achieves **85% task success and mean episode length 856 steps**. Behavioral cloning yields $R^2 = 0.872$ and **55/56 terms** at $\lambda = 0.05$, comparable to the standard formulation's 82/84 at $\lambda = 0.10$, but at 5 percentage points lower neural-policy success (85% vs. 90%). The no-x variant does not outperform the standard formulation: both achieve comparable distillation sparsity while the standard formulation achieves higher task success. This suggests the degree-3 polynomial basis already captures the translational symmetry implicitly through the relationship between x, $\dot{x}$, and the dynamics, making the explicit coordinate removal redundant. We also examined SE(3) forward-kinematics coordinates (replacing raw angles with absolute-link sin/cos), which produced catastrophic ill-conditioning ($\kappa = 10^{17}$–10^{19}) due to unit-circle constraints ($\sin^2\theta + \cos^2\theta = 1$) creating near-exact linear dependence in the degree-2+ polynomial features — confirming that the choice of coordinate representation has a larger effect on numerical stability than on task performance, and that degree-3 in raw angles is difficult to improve upon at this data scale.

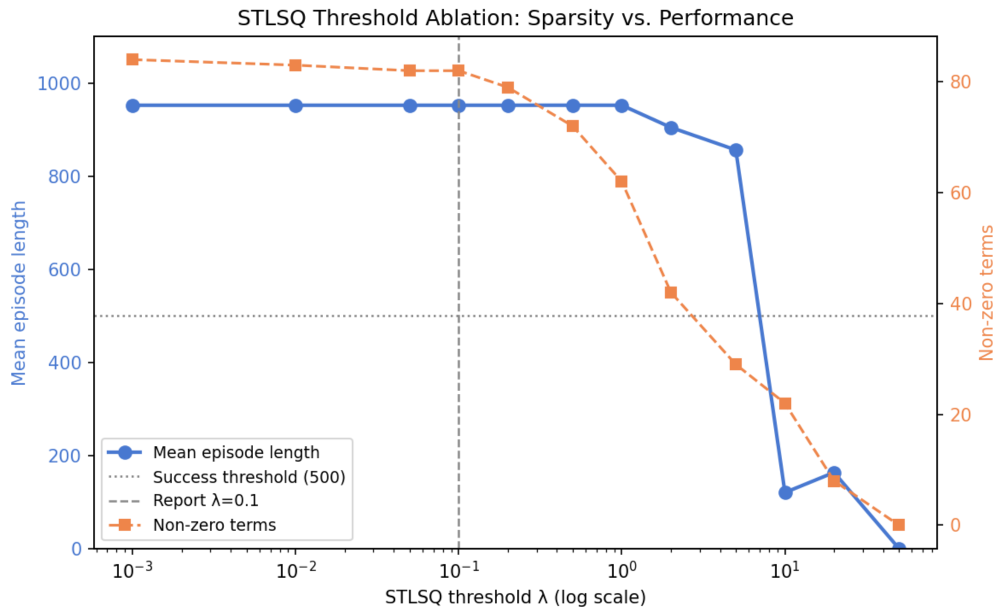

Figure 6. STLSQ threshold ablation. Left axis: mean episode length (blue); right axis: non-zero term count (orange). Dashed vertical line marks the report threshold ($\lambda = 0.10$). Performance is stable at 90–95% success across the full ablation range.

4.6 Algorithm and Library Variants

Algorithm 1 treats the feature library Θ and the policy optimizer A as explicit practitioner inputs. The strict full-horizon test (§4.2) established that pushing PPO+polynomial harder — 323,934 real steps at 5× the original budget — cannot close the gap: data efficiency collapsed without improving the result. The bottleneck is not iteration count but the choice of A and Θ. We evaluated four variants to characterize the design space.

PPO + no-x (feature ablation). Removing cart position x from the SINDy library exploits the IDP's translational symmetry ($\Delta x = \dot{x}\,\Delta t$ exactly by kinematics), reducing the degree-3 library from 120 to 84 features. Convergence required 35,111 real steps (**11.4**×) at 85% success. Distillation (5D policy input, 56-feature library) yielded 55/56 terms — still essentially dense — confirming the polynomial ill-conditioning is not resolved by removing a single coordinate.

SAC + polynomial (algorithm swap only). Replacing PPO with Soft Actor-Critic converged in 30,735 real steps (**13.0**×) with **100% NN task success**. SAC's off-policy replay buffer is theoretically a liability in Dyna-style MBRL because the surrogate is refitted each iteration; in practice, entropy regularization appears to compensate on this system. The distilled polynomial remains near-dense (83/84 terms), confirming that ill-conditioning is independent of the optimizer choice.

PPO + Lagrangian (library swap only). Replacing the polynomial library with 32 atoms derived from the IDP Euler-Lagrange equations — kinetic energy terms, gravitational restoring forces, Coriolis interactions, and cart–pole coupling — changes the representational structure fundamentally. Each atom is physically meaningful by construction, so ridge regression replaces STLSQ for dynamics fitting (all atoms are genuinely present in the IDP dynamics; thresholding would incorrectly zero real physical contributions). A kinematic fidelity check confirms the recovered coefficients reproduce $\Delta x = \dot{x}\,\Delta t$ to within 1% error. Convergence required 25,984 real steps (**15.4**×) at **100% NN task success**.

SAC + Lagrangian (both swaps). Combining both changes yields the strongest result. The Dyna loop converges in two iterations using **22,723 real steps — 17.6× fewer than the full-order baseline** — with 100% task success and mean episode length of 1,000 steps, matching the full-order performance ceiling. The Lagrangian surrogate has higher next-state RMSE than the polynomial surrogate — it only models physically relevant dynamics — yet achieves substantially better closed-loop performance, confirming that RMSE on generic next-state prediction is a poor quality metric for physics-informed surrogates. The distilled Lagrangian controller (29 terms, $R^2 = 0.958$) achieves 100% task success ($\geq$ 500 steps) at mean episode length 715. Every retained term is physically interpretable: velocity damping, gravitational restoring forces, inter-link Coriolis coupling.

Variant	Real-env steps	Efficiency	NN success	Distilled terms	Distilled success
Baseline PPO	400,000	1.0×	100%	—	—
PPO + poly (strict)‡	323,934	1.2×	95%	84/84 (dense)	97%‡
PPO + polynomial	66,277	6.0×	90%	82/84 (dense)	85%
PPO + no-x	35,111	11.4×	85%	55/56 (dense)	85%
SAC + polynomial	30,735	13.0×	100%	83/84 (dense)	70%
PPO + Lagrangian	25,984	15.4×	100%	29/29 (interp.)	100%
SAC + Lagrangian	**22,723**	**17.6×**	**100%**	**29/29 (interp.)**	**100%**

‡Strict continuation targeting full-horizon (≥999-step) success; 95% at the standard ≥500-step criterion. Peak at the harder criterion was 80% after 108,251 steps; the loop continued to 323,934 steps without improvement. Distilled success (97%) also evaluated at the ≥999-step criterion.

The SAC swap improves data efficiency; the Lagrangian swap delivers interpretability; both together achieve all three goals simultaneously.

4.7 Code Repository

All code and results: **https://github.com/falconeaj1/ME_595**. Key notebooks: `full-order-simulation.ipynb` (baseline PPO), `sindy-rl.ipynb` (SINDy-RL pipeline), and `sindy-rl-no-x.ipynb` (translational-symmetry variant). Professor Michelle Hickner added as collaborator (GitHub: mhickner).

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5 Summary

We stress-tested SINDy-RL against its three core goals on the inverted double pendulum. **SINDy-RL passes the stress test.** With practitioner-appropriate choices of feature library and policy optimizer — both explicit inputs to Algorithm 1 — the algorithm simultaneously achieves data economy, a reduced-order policy, and an interpretable controller on a genuinely hard unstable benchmark.

Goal 1 (data economy) is confirmed across all variants. Every library–optimizer combination substantially outperforms the full-order PPO baseline, from 6.0× (PPO+polynomial) to 17.6× (SAC+Lagrangian). Pushing PPO+polynomial harder — a stricter convergence criterion consuming 323,934 real steps — confirms that the efficiency gains are structural to the Dyna architecture and cannot be improved by brute-force iteration alone.

Goal 2 (reduced-order policy) is confirmed. The SAC+Lagrangian distilled controller is 29 terms — 336× smaller than the baseline NN — with 100% task success. The PPO+polynomial controller (82 terms, 119×, 85% success) also confirms the reduced-order goal.

Goal 3 (interpretability) requires a physics-informed library. The degree-3 polynomial basis is ill-conditioned on the IDP ($\kappa \approx 2.4 \times 10^4$): STLSQ retains 82 of 84 terms regardless of threshold, and a threshold sweep confirms this cannot be tuned away. This is a finding about representation selection, not a limitation of SINDy-RL. The 32-atom Lagrangian library delivers interpretability by construction — every retained term is a physically meaningful quantity from the IDP equations of motion — because the basis encodes what the system actually does rather than generic polynomial monomials.

Reaching any result required resolving two non-obvious engineering obstacles absent from the algorithm description: the degree-2 RMSE ceiling (a degree-2 library cannot express the IDP's cubic inter-modal coupling terms — fix: degree-3), and surrogate exploitation (ensemble uncertainty penalization cannot detect shared extrapolation errors on a shared polynomial basis — fix: uncertainty penalty plus real-environment rollback together). Both lessons generalize: when RMSE does not decrease with $10\times$ more data, the cause is model capacity; when an ensemble shares a function basis, only real-environment feedback reliably detects out-of-distribution extrapolation.

The key practical finding is that library selection dominates outcome. A generic polynomial library is flexible but ill-conditioned; a physics-informed Lagrangian library reduces feature count, improves conditioning, and delivers interpretability as a structural property. For any mechanical system with a known Lagrangian, the Lagrangian library is the appropriate choice of Θ in Algorithm 1.

CRediT Statement (https://credit.niso.org)

Role	Patrick Smith	Andrew Falcone
Conceptualization	Yes	Yes
Data curation	Yes	Yes
Formal analysis	Lead	Supporting
Investigation	Yes	Yes
Methodology	Lead	Supporting
Software	Lead	Supporting
Validation	Yes	Yes
Visualization	Yes	Yes
Writing – original draft	Yes	Yes
Writing – review & editing	Yes	Yes

AI tool disclosure: Claude (Anthropic) assisted with code drafting, debugging, writing iteration, and figure generation. All analysis, results, and conclusions were reviewed and executed by the authors, who take full responsibility for the submitted work.

References

[1] N. Zolman, C. Lagemann, U. Fasel, J. N. Kutz, and S. L. Brunton, "SINDy-RL: Interpretable and efficient model-based reinforcement learning," *Nature Communications*, 2025.

[2] A. B. Arrieta *et al.*, "Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI," *Information Fusion*, vol. 58, pp. 82–115, 2020.

[3] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Machine Intelligence*, vol. 1, no. 5, pp. 206–215, 2019.

[4] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proceedings of the National Academy of Sciences*, vol. 113, no. 15, pp. 3932–3937, 2016.

[5] R. S. Sutton, "Integrated architectures for learning, planning, and reacting based on approximating dynamic programming," in *International conference on machine learning (ICML)*, 1990, pp. 216–224.

[6] E. Kaiser, J. N. Kutz, and S. L. Brunton, "Sparse identification of nonlinear dynamics for model predictive control in the low-data limit," *Proceedings of the Royal Society A*, vol. 474, no. 2219, p. 20180335, 2018.

[7] U. Fasel, J. N. Kutz, B. W. Brunton, and S. L. Brunton, "Ensemble-SINDy: Robust sparse model discovery in the low-data, high-noise limit," *Proceedings of the Royal Society A*, vol. 478, no. 2260, 2022.

[8] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms." 2017.

[9] M. R. Schroeder, "Synthesis of low-peak-factor signals and binary sequences with low autocorrelation," *IEEE Transactions on Information Theory*, vol. 16, no. 1, pp. 85–89, 1970.

[10] S. Ross, G. Gordon, and D. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," in *AISTATS*, in PMLR, vol. 15. 2011, pp. 627–635.

[11] B. M. de Silva, K. Champion, M. Quade, J.-C. Loiseau, J. N. Kutz, and S. L. Brunton, "PySINDy: A Python package for the sparse identification of nonlinear dynamical systems from data," *Journal of Open Source Software*, vol. 5, no. 49, p. 2104, 2020.

[12] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-Baselines3: Reliable reinforcement learning implementations," *Journal of Machine Learning Research*, vol. 22, no. 268, pp. 1–8, 2021.

[13] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *IEEE/RSJ international conference on intelligent robots and systems (IROS)*, 2012, pp. 5026–5033.